The Clebsch hexagon code: rigidity of a conic syndrome locus

An MDS code over $\mathbb{F}_{11}$ whose projective deep-hole directions form a conic

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Working draft — July 2026

Abstract

Among the six-arcs of $\mathrm{PG}(2,11)$, exactly one projective-equivalence class has the maximum-distance syndrome directions of its associated MDS code contained in a conic. It is the *Clebsch hexagon*, and this coding-theoretic condition alone recovers its full projective stabilizer A_5. The characterization remains true if "conic" is weakened to the zero set of any nonzero homogeneous form of degree at most three.

The arc itself lies on no conic, so the associated $[6,3,4]_{11}$ MDS code is non-GRS; its covering radius is three. Its twelve projective deep-hole directions are exactly the $\mathbb{F}_{11}$-rational points of the conic. They lift to 120 deep-hole cosets, 2400 minimum-weight leaders, and 159720 received-word deep holes; the cosets form one monomial orbit, and the received words form one orbit under monomial automorphisms and codeword translations. A single quadratic in the syndrome gives a complete distance oracle, while the radius-two ambiguity pattern reconstructs the classical Brianchon points and self-polar triangles.

We prove the rigidity as a five-way equivalence and quantify it by a sharp nearest-conic gap of twelve. We also decompose the 252 one-point perturbations into eight A_5-orbits and construct an automorphism-invariant unordered bipartition of the twenty weight-three support types, giving a $10+10$ split of the leaders in every deep-hole coset. For six-arcs, a chord-multiplicity identity and the clique number of the exact-distance-two graph of the Sylvester graph show that $q=11$ is the only conic-filling field. More broadly, for arc sizes four through seven, the only conic-filling cases are a four-frame in $\mathrm{PG}(2,5)$ and the Clebsch hexagon. The six-arc census is classical; our contribution is the rigidity theorem, its quantitative and decoding refinements, and their coding-theoretic interpretation.

1 Introduction

Let C be a linear $[n,k,d]_q$ code with covering radius ρ and full-rank parity-check matrix H. For a received word v, write $\sigma(v)=Hv^{\mathsf{T}}$ and

$$w(s)=\min\{\mathrm{wt}(e):He^{\mathsf{T}}=s\}.$$

Then $d(v,C)=w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ. The syndrome s specifies one coset of C, whose coset leaders are its weight-$w(s)$ representatives. Because $w(\lambda s)=w(s)$ for $\lambda\neq 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\mathrm{proj}}(C)=\{[s]\in\mathrm{PG}(n-k-1,q):w(s)=\rho\}.$$

Thus one projective direction represents $q-1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r=n-k$, the parity-check columns form a projective arc in $\mathrm{PG}(r-1,q)$, and $w(s)$ is the least number of columns whose span contains

$[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by $r-1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. This arc–coset dictionary, in various forms, underlies the classical theory of complete arcs and saturating sets and has been made precise and general for MDS codes by Davydov, Marcugini, and Pambianco [7], whose Theorem 6.3 gives the exact leader-count formula we use in Section 3. For plane parity-check systems — equivalently redundancy-three $[n, n-3, 4]_q$ MDS codes — the same dictionary is implicit already in the classical arc/covering-radius literature; see Hirschfeld [17]. We write $\mathcal{U}(A)$ for this projective extension locus, the directions off A and off all its secants. It is the weight-three syndrome-direction locus. When $\mathcal{U}(A) \neq \emptyset$ the associated redundancy-three code has $\rho = 3$ and $\mathcal{U}(A) = \mathcal{D}_{\mathrm{proj}}(C)$; when $\mathcal{U}(A) = \emptyset$ the covering radius is at most two, and its deep-hole directions are therefore not described by $\mathcal{U}(A)$. For the standard (affine) Reed–Solomon setting, determining deep holes is a long-studied algorithmic and structural question; Guruswami and Vardy [16] initiated its decision-complexity study, showing maximum-likelihood decoding of Reed–Solomon codes to be NP-hard by essentially proving that deciding whether a received word is a deep hole is co-NP-complete, and Cheng and Murray [6] gave a substantially simpler proof of that same co-NP-completeness result and extended the study to standard (full-evaluation-set) Reed–Solomon codes. Zhang, Wan, and Kaipa [28] determine the deep holes of *projective* Reed–Solomon codes completely, identifying them with tangent lines to, and a quadratic-extension family over, the underlying normal rational curve (their Theorems I.4–I.7). Both of these are for the generalized Reed–Solomon (GRS) case, where the arc lies on a normal rational curve; the present code does not.

The object of this paper is a single, small, and highly symmetric exception to that GRS world: a $[6, 3, 4]_{11}$ MDS code not monomially equivalent to a GRS code, whose covering radius is three and whose projective deep-hole syndrome locus — the directions missed by the arc's fifteen secants — coincides exactly with the $\mathbb{F}_{11}$-points of a single nonsingular conic. The arc realizing this code is the *Clebsch hexagon*: six points of $\mathrm{PG}(2, 11)$. Its real-projective antecedent goes back to Clebsch [3, p. 336]; Edge constructed the finite-plane configuration, named it a Clebsch hexagon, and determined its order-60 stabilizer [4, §§29–32]. Dye developed its synthetic geometry over a general field, proved its projective uniqueness, and determined its stabilizer [14, Theorems 1 and 3, pp. 275–278]; Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [25].

That such an arc exists, and that its projective deep-hole syndrome locus is a conic, is by itself a short synthesis of classical geometry and the DMP dictionary, independently certified by finite computation. What we add is that this instance is not an accident of the arc chosen: among *all* six-arcs of $\mathrm{PG}(2, 11)$ — fifteen classes under $\mathrm{PGL}(3, 11)$, which our search meets as 1548 frame-normalized representatives — the Clebsch hexagon is the unique class whose projective deep-hole syndrome locus lies on any conic at all, degenerate or not, and this single containment condition already forces the stabilizer to contain A_5. We state this as a rigidity theorem (Section 4), quantify it by a global nearest-conic gap and an orbit-resolved local perturbation theorem, exhibit an invariant chirality bipartition of the weight-three support types and its induced per-coset leader split (Section 5), and prove that $q = 11$ is the only prime power at which the conic-filling extension locus can occur (Section 6). The proof combines a universal chord-defect identity with the Sylvester graph on the internal points of a conic in $\mathrm{PG}(2, 9)$; it assumes no symmetry of the arc. The previous nine-field census is retained as independent exact verification. For the classical Clebsch family itself, the same identity and Dye's ten Brianchon points give the all-field formula $|\mathcal{U}(H)| = q^2 - 14q + 45$; when $q \equiv 3 \pmod 4$, the associated conic lies inside this locus with exact excess $(q-4)(q-11)$.

What is new and what is not. The six-arc itself, its real-projective ancestry, and its finite conic geometry are classical: Clebsch [3], Edge [4], and Dye [14]. Projective uniqueness already follows from Dye's Theorem 1; Storme and Van Maldeghem independently place the arc in the finite-plane catalogue and establish the incompleteness statement used here [25]. In the later terminology of Blokhuis, Seress, and Wilbrink, the same configuration is a complete exterior set of a conic [2, pp. 143, 146]; their broader sets-without-tangents work and Van de Voorde's account give the adjacent literature [1, 26]. The six-arc census in $\mathrm{PG}(2,11)$ is due to Sadeh [18, 22]; see also the tables in Hirschfeld's book [17]. We recompute its extension-count spectrum as a byproduct of the rigidity theorem's exhaustive search and claim no priority on the raw numbers. The identification of this code's projective deep-hole syndrome locus with the A_5-invariant conic (Proposition 3.2) is recorded, with a kernel-checked certificate, in the companion arcs paper [5, Prop. 8.7]; we reprove it here to keep the present paper self-contained and claim no priority for it either. What we claim as new is the *reading* of that geometry: that the conic-containment condition characterizes the Clebsch hexagon among all six-arcs and recovers A_5 from a purely coding-theoretic hypothesis, not by construction; the global and local quantitative gap theorem accompanying that rigidity; the invariant chirality bipartition of the weight-three support types and its induced per-coset leader split; and an explicit decoder-side synthesis in which a four-way syndrome oracle and its ambiguity data reconstruct the classical Brianchon configuration; and the unconditional finite-field argument isolating $q=11$, including the Clebsch-family uncovered formula obtained by combining Dye's classical geometry with the chord-defect identity.

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C}:\ XZ = Y^2$$

in $\mathrm{PG}(2,11)$, with its 12 points identified with $\mathrm{PG}(1,11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}(3,11)$ is $\mathrm{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

Definition 2.1 (Clebsch hexagon)**.** Let $A_5 \subset \mathrm{PGL}_2(11)$ be an icosahedral subgroup. The *Clebsch hexagon* is the six-point orbit of poles of the six 5-fold-axis chords of A_5 acting on $\mathcal{C}$; equivalently, the six points of $\mathrm{PG}(2,11)$ each of whose two tangency lines to $\mathcal{C}$ meet $\mathcal{C}$ in an antipodal pair fixed by a Sylow 5-subgroup of A_5.

For a nonsingular conic over an odd field, an *external point* lies on two tangents, while an *external line* is disjoint from the conic. Edge constructs exactly this $q=11$ hexagon: its six vertices are external points, all fifteen joins are external lines, and the joins concur three at a time at ten internal Brianchon points [4, §§29–32]. In the terminology of Blokhuis, Seress, and Wilbrink, it is the six-point complete exterior set of $\mathcal{C}$ [2, pp. 143, 146]; see Van de Voorde [26] for the later exterior-set context. This exterior-set condition says only that the joins avoid $\mathcal{C}$. Proposition 3.2 is the strictly stronger covering statement that those joins cover every point outside the hexagon and $\mathcal{C}$. Van de Voorde also relates the broader sets-without-tangents problem to stopping sets in projective-plane LDPC codes; that is a different coding connection from the MDS covering-radius and deep-hole interpretation developed here.

The arc hypothesis is essential even inside this classical exterior-set category. Blokhuis–Seress–Wilbrink's exhaustive $q=11$ list has two complete exterior configurations: the

Clebsch six-arc and a Pasch configuration [2, p. 146]. The latter has four collinear triples on its six points, so it cannot be the column set of a redundancy-three MDS code. Thus complete exteriority by itself gives only the inclusion $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; imposing the arc/MDS condition selects the Clebsch branch on which the exact covering and rigidity questions of this paper are posed.

There is also a more direct fixed-conic binary-code lineage. Droms, Mellinger, and Meyer formed LDPC codes from the four internal/external and passant/secant point–line incidence matrices [15]; two of their dimension conjectures—the secant–external and passant–internal cases—were subsequently proved in [23, 20]. Wu then used whole $\mathrm{PSL}_2(q)$-orbit families of conics contained in the internal-point set, and Madison and Wu treated the external-point analogue [27, 21]. These are binary incidence codes whose coordinates range over an entire internal or external point class. For example, at $q = 11$ the passant–internal null-space code has parameters $[55, 25]_2$, while Wu's internal-orbit-conic null-space code has parameters $[55, 31]_2$. They are therefore different code objects from the six-coordinate $\mathbb{F}_{11}$-linear MDS code studied here; the shared conic, polarity, and $\mathrm{PSL}_2(11)$ geometry is infrastructure rather than a competing covering-radius or deep-hole identification.

This is projectively the arc K_2 of Storme and Van Maldeghem [25, Prop. 11] at $q = 11$, and the hexagon studied synthetically by Dye under the existence condition 5 a square in the base field [14, Theorem 1]. Dye's Theorems 1 and 3 give projective uniqueness and stabilizer A_5 directly; Storme and Van Maldeghem's Proposition 12 gives the corresponding finite-plane catalogue statement. Dye's unique-polarity theorem identifies an associated conic; in our chosen coordinates the directly verified conic is $\mathcal{C}$. His calculation also shows that the six vertices lie on no conic in characteristic 11 [14, Theorem 2, pp. 277–278; p. 281].[1] The arcs of $\mathrm{PG}(n, q)$ fixed by a *primitive* A_5 or A_6 were catalogued more generally by O'Keefe and Storme [19], of which the present arc is the planar A_5 case (its stabilizer is 2-transitive on the six points, hence primitive); we cite that catalogue for completeness, and take the planar uniqueness statement we use from [25, Prop. 12]. Storme and Van Maldeghem's Proposition 13 records, by computer search, that this six-arc is incomplete at $q = 11$ (unlike at $q = 9$, where the analogous A_5-hexagon is complete); equivalently, its associated redundancy-three code has covering radius exactly three. Incompleteness gives the lower bound, and the three-dimensional syndrome space gives the upper bound.

We work throughout with the explicit representative

$$A = \{(1, 10, 0),\ (1, 9, 1),\ (1, 4, 7),\ (1, 8, 5),\ (0, 1, 4),\ (1, 1, 7)\},$$

already used in the companion arcs manuscript [5] for the same reasons: it is disjoint from $\mathcal{C}$, no three of its points are collinear, and its stabilizer in $\mathrm{PGL}(3, 11)$ has order 60.

3 The code and its projective syndrome locus

Form the 3×6 matrix H whose columns are (representatives of) the six points of A, and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^{\mathsf{T}} = 0\}$$

be the resulting $[6, 3, 4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is

[1] The primary texts separate two statements that are easy to conflate. Dye proves that, over $\mathbb{F}_{11}$, every edge is external to the associated conic [14, discussion preceding Theorem 6, p. 281], and Blokhuis–Seress–Wilbrink call the resulting six-arc a complete exterior set; thus the conic is contained in the uncovered locus. Neither source states the converse covering assertion that every off-conic point lies on an edge, which is the equality in Proposition 3.2.

nonsingular), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [24].

By the arc–coset dictionary [7, Def. 6.1, Thm. 6.3], for every nonzero syndrome s above a projective direction $x = [s]$, the corresponding coset has minimum weight one, two, or three according as $x \in A$, x lies on a secant of A, or x lies on neither. In the last case each of the $q-1$ cosets above x has exactly $\binom{n}{3}$ weight-three leaders, here $\binom{6}{3} = 20$. Their Theorem 6.3 is stated for an arbitrary n-arc of $\mathrm{PG}(2, q)$ and its associated $[n, n-3, 4]_q$ MDS code, with no Reed–Solomon, normal-rational-curve, or conic hypothesis and no restriction on q or the characteristic; it therefore applies to the non-GRS code C at hand. (The GRS and conic specializations appear separately in their Theorems 6.4, 6.5, 6.8, and 6.10, which we do not use. The leader count is also the $d = 4$, $n = 6$ case of the general $\binom{n}{d-1}$ of [7, Thm. 7.7], so it is an instantiation of a uniform formula rather than a coincidence at these parameters.) Finally, by their Definition 6.1 together with the definition of arc completeness, adjoining x as a seventh arc column preserves the MDS property exactly when x is missed by every secant.

The stabilizer already forces the small point sets that will occur.

Proposition 3.1 (The A_5 point orbits)**.** *Let $G = \mathrm{Stab}(A) \cong A_5$. Its orbits on $\mathrm{PG}(2, 11)$ have lengths*

$$6,\ 10,\ 12,\ 15,\ 30,\ 30,\ 30.$$

The first four are respectively A, the ten Brianchon points, $\mathcal{C}(\mathbb{F}_{11})$, and the fifteen vertices of the five self-polar triangles. In particular, $\mathcal{C}(\mathbb{F}_{11})$ is the unique twelve-point G-orbit.

Proof. Dye's polarity and stabilizer theorems identify G with the ternary icosahedral representation preserving $\mathcal{C}$, and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [14, Theorems 2–3 and Corollary 1, pp. 276–279]. For completeness, we derive the orbit sizes rather than import them from the finite enumeration.

Since $11 \nmid 60$, the representation is semisimple, and the ternary icosahedral representation is irreducible; hence G fixes no projective point. The ordinary three-dimensional icosahedral character, reduced in $\mathbb{F}_{11}$, shows that an involution, an element of order three, and an element of order five fix respectively 13, 1, and 3 projective points. Indeed, an involution has a one-dimensional eigenspace and a two-dimensional eigenspace; the nontrivial order-three eigenvalues are not in $\mathbb{F}_{11}$; and the three order-five eigenvalues are distinct because $5 \mid 10$. Restricting the same representation to the standard subgroups of A_5 gives the following common-fixed-point ledger; the standard subgroup classification of A_5 shows that these are all possible nontrivial proper point stabilizers. Here D_5 has order ten, and the last column counts points whose stabilizer is exactly the displayed subgroup type.

H	A_4	D_5	S_3	C_5	V_4	C_3
$\#H$	5	6	10	6	5	10
$\lvert \mathrm{Fix}_{\mathrm{PG}}(H) \rvert$	0	1	1	3	3	1
exact-stabilizer points	0	6	10	12	15	0

To spell out the subtraction in the last row: the normalizer D_5 fixes one of the three C_5 eigenlines and interchanges the other two, giving 6 points with stabilizer D_5 and 12 with stabilizer C_5; the unique C_3-fixed line is already fixed by its S_3 normalizer; and A_4 cycles the three V_4 eigenlines, leaving all fifteen with exact stabilizer V_4.

It remains to account for exact C_2 stabilizers. Each involution lies in two D_5 subgroups, two S_3 subgroups, and one V_4. Of its thirteen fixed points, these contribute respectively 2, 2, and 3 points from the preceding rows. The remaining six points have exact stabilizer C_2. There are fifteen involutions, so this gives 90 points, necessarily three orbits of length $60/2 = 30$. The total $6 + 10 + 12 + 15 + 90 = 133$ exhausts the plane. Dye's Theorem 6(v), specialized at $q = 11 \equiv 1 \pmod{10}$, identifies the twelve-point C_5-stabilizer orbit with $\mathcal{C}(\mathbb{F}_{11})$ [14, pp. 281–282; §3.1, p. 284]; the other three identifications are the transitive sets cited above. $\square$

The explicit sixty-element action, all seven orbits, the unique six- and twelve-orbits, and the order-five fixed-point union $A \cup \mathcal{C}(\mathbb{F}_{11})$ are independently kernel-checked by `lean/RelativeConicArcs/Q11A5PointOrbits.lean`. This finite certificate verifies the displayed concrete action, not the abstract subgroup and character derivation in the proof.

Proposition 3.2 (The projective deep-hole syndrome locus is the conic). *The covering radius of C is three, and its projective deep-hole syndrome locus $\mathcal{D}_{\text{proj}}(C)$ — the set of directions $x \notin A$ on no secant of A — is exactly the twelve points of $\mathcal{C}$.*

Proof. The set $\mathcal{U}(A)$ is G-invariant. Dye's Clebsch classification gives exactly ten Brianchon points. For completeness, here is the relevant double count. The fifteen edges contribute 150 off-arc incidences. The 45 pairs of disjoint edges meet once, so if n_i counts off-arc points lying on exactly i edges, then

$$n_1 + 2n_2 + 3n_3 = 150, \qquad n_2 + 3n_3 = 45, \qquad n_3 = 10.$$

Thus the edges cover $n_1 + n_2 + n_3 = 115$ of the 127 off-arc points, leaving $|\mathcal{U}(A)| = 12$. This is the $q = 11$ specialization of the chord-defect identity in Lemma 6.2. By definition $\mathcal{U}(A)$ is disjoint from the unique six-point orbit A. The orbit lengths in Proposition 3.1 therefore force this invariant twelve-set to be the unique twelve-point orbit $\mathcal{C}(\mathbb{F}_{11})$. By the dictionary above this is exactly the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three rather than two. Dye's edge criterion and the complete-exterior-set description of Blokhuis–Seress–Wilbrink independently supply the classical inclusion $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$ [14, discussion preceding Theorem 6, p. 281] [2, pp. 143, 146]. $\square$

The conclusion and full 133-point incidence table are reproduced independently by `check_rigidity_degenerate_conic.py` as the $|\mathcal{U}(A)| = 12$ entry of the census of Section 4.

This identification, together with its coset data and a kernel-checked Lean certificate (root module `lean/RelativeConicArcs/Q11Coding.lean`), is also recorded in the companion arcs paper [5, Prop. 8.7], where it arises as one instance of that paper's secant-index spectrum; we restate it here, with a self-contained proof, because it is the starting point of the rigidity theorem that is the present paper's subject. The results of Sections 4–6 — the rigidity equivalence, the gap theorem, the chirality invariant, and the isolation of $q = 11$ — do not appear in that companion paper; their broader priority is treated separately in the literature comparison of Section 4.

Corollary 3.3 (Directions, cosets, leaders, and received words). *The projective deep-hole syndrome locus has* 12 *directions, exactly the points of $\mathcal{C}$. Above them lie* $12(11-1) = 120$ *nonzero maximum-distance syndromes, hence* 120 *deep-hole cosets. Each such coset has* $\binom{6}{3} = 20$ *minimum-weight leaders, for* 2400 *leaders in all, and contains* $|C| = 11^3 = 1331$ *received words. Consequently C has* $120 \cdot 1331 = 159720$ *received-word deep holes.*

Proof. Immediate from Proposition 3.2, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \to \mathrm{PG}(2,11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. $\square$

Proposition 3.4 (One orbit on received-word deep holes)**.** *The monomial automorphism group of C is transitive on the 120 deep-hole cosets. Let Γ be the group generated by these automorphisms and by translations $v \mapsto v + c$ for $c \in C$. Then*

$$|\Gamma| = |C|\,|\,\mathrm{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ is transitive on all 159720 received-word deep holes. The stabilizer of each such word is cyclic of order five.

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six axis chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^\times$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w, choose $m \in \mathrm{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w. Both kinds of generator are Hamming isometries preserving C. The monomial group normalizes the translation subgroup, since $m\,t_c\,m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \mathrm{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. $\square$

As an independent finite verification, `check_chirality.py` constructs $C = \ker H$, verifies all $600 \cdot 1331$ monomial images of its codewords, and certifies the resulting 159720-element orbit.

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight $0, 1, 2, 3$, summing to $11^3 = 1331$); in particular the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters; we record it as a sanity check, not a novelty.

The special geometry also makes distance diagnosis and nearest-word structure unusually explicit. Write a syndrome as $s = (s_0, s_1, s_2)$ and put $Q(s) = s_0 s_2 - s_1^2$.

Proposition 3.5 (Complete syndrome oracle and ambiguity enumerator)**.** *For $v \in \mathbb{F}_{11}^6$ with syndrome $s = Hv^{\mathsf{T}}$,*

$$d(v, C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \text{ and } Q(s) = 0, \\ 2, & \text{otherwise.} \end{cases}$$

Among the 1330 nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are

$$960, \qquad 150, \qquad 100, \qquad 120.$$

For a distance-two syndrome the number of nearest codewords is its secant index. Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.

Proof. The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 3.2 identifies the last set with $Q = 0$, proving the oracle. In the notation of its proof, the chord-intersection equations give $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$, where n_i counts projective directions of secant index i and n_0 is the uncovered locus. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders. □

For this fixed six-coordinate code, the proposition is a constant-size post-syndrome test: it is not an asymptotic complexity claim. The complete coefficient-bearing enumeration is independently replayed by `check_decoding.py`; the finite oracle, uniform twenty-support theorem, and ambiguity counts are also packaged in `Q11DecodingSynthesis.lean` under `lean/RelativeConicArcs/`.

3.1 Certified structural facts

The following facts are either forced by the MDS parameters or are direct consequences of the arc's known geometry; we record them because they anchor the code's symmetry and design-theoretic reading, not as new results.

- **Automorphism groups.** The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the six coordinate supports in its exotic degree-six action. It is the support-permutation quotient of the monomial automorphism group, which fits into the central exact sequence

$$1 \longrightarrow \mathbb{F}_{11}^{\times} \longrightarrow \mathrm{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

  The kernel consists of the ten global coordinate scalars. The sequence splits: since $\gcd(3, 10) = 1$, every class in $\mathrm{PGL}(3, 11)$ has a unique determinant-one representative, and these representatives multiply; their associated monomial lifts give an A_5 complement. Thus $\mathrm{MAut}(C) \cong C_{10} \times A_5$ has order 600. By contrast, for the displayed choice of column representatives the subgroup of pure coordinate permutations is trivial. The exact support image, scalar kernel, monomial order, pure subgroup, and transitivity on the 120 nonzero syndromes above $\mathcal{C}$ are independently checked by `check_code_automorphisms.py`; Proposition 3.2 identifies these with the maximum-distance syndromes.

- **Design view.** The MDS parameters make C an orthogonal array $\mathrm{OA}(11^3, 6, 11, 3)$ of strength three; this is an automatic consequence of the $[6, 3, 4]$ parameters and is recorded here only to connect the code to the design-theoretic framing this venue expects.

- **Five self-polar triangles.** Edge's finite-plane construction [4, §§30.2–31] and Dye's general-field synthetic theory [14, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, each self-polar with respect to the unique conic $\mathcal{C}$ stabilized by the hexagon's projective stabilizer — the same conic as the projective deep-hole syndrome locus of Proposition 3.2. In the six-coordinate labelling, the three vertex-pairs underlying each triangle form a perfect matching, and the five matchings partition all fifteen pairs: they are a synthematic total. Pairing two of these triangles produces an alternating six-cycle. Its two colour classes form a complementary pair of three-element coordinate supports, while its opposite-vertex matching gives one of the ten Brianchon concurrences. Thus the ten unordered pairs of triangles index simultaneously the ten Brianchon points and the ten complementary support pairs; this is the geometric origin of the Petersen graph made precise in Proposition 5.1.

4 The rigidity theorem

The content of Proposition 3.2 is, on its own, a short synthesis for one known arc, independently certified by finite computation. The following theorem is the paper's main claim: among *all* six-arcs of $\mathrm{PG}(2,11)$, the property "the projective deep-hole syndrome locus lies on a conic" picks out the Clebsch hexagon uniquely, and does so in a way that recovers its projective stabilizer from the coding condition rather than assuming it.

Remark 4.1 (Two conditions, one word apart). The condition studied here is that $\mathcal{U}(A)$ — the projective uncovered, or projective deep-hole syndrome, locus — lies on a conic. It is *not* the familiar condition that the arc A itself lies on a conic, which is the standing hypothesis throughout the arc literature and which the Clebsch hexagon famously fails. The two are logically unrelated: they constrain different point sets, and the arcs satisfying them are disjoint. Indeed every six-arc lying *on* a conic in $\mathrm{PG}(2,11)$ has $|\mathcal{U}(A)| \in \{18, 19, 20, 22\}$, so by (iii) none of them comes anywhere near the condition of the theorem.

Census framing. The projective-equivalence classes of k-arcs in $\mathrm{PG}(2,11)$ were classified by Sadeh [22, 18]; see also Hirschfeld [17, Ch. 14]. Over the six-arc classes of that census we compute, for each representative A, the projective uncovered locus $\mathcal{U}(A)$ — the points off A and off all fifteen of its secants, equivalently the points extending A to a projective seven-arc — and read $\mathcal{U}(A)$ as the projective covering-radius-three syndrome data of the associated $[6,3,4]_{11}$ code. *We claim no priority on the six-arc census, nor on the raw extension-count data* $|\mathcal{U}(A)|$ *that an extension-based classification entails*: the size of $\mathcal{U}(A)$ is a standard projective invariant of any arc, and the census itself is Sadeh's. What we contribute is the geometric and coding-theoretic meaning of $\mathcal{U}(A)$: that its containment in a conic singles out the Clebsch hexagon and forces A_5, that this containment is rigid rather than merely typical, and the resulting gap, decoding, and support structure. The exact conic locus for the singled-out class is the setup supplied by Proposition 3.2, not a separate novelty claim here.

We first isolate the geometric input that removes degenerate conics from the computer-assisted part of the argument.

Lemma 4.2 (Six-arc line bound)**.** *Let A be a six-arc in* $\mathrm{PG}(2,q)$*, where q is odd. Every line contains at most $q-5$ points of* $\mathcal{U}(A)$*.*

Proof. A chord contains no point of $\mathcal{U}(A)$. If a non-chord line ℓ contains one vertex, the ten chords on the other five vertices meet ℓ in at least five distinct points: their intersections give a proper edge-colouring of K_5, whose chromatic index is five. Together with the vertex, at least six of the $q+1$ points of ℓ are therefore outside $\mathcal{U}(A)$.

It remains to consider $\ell \cap A = \varnothing$. The fifteen chords meet ℓ in covered points, and at most three chords pass through any one of them, because their endpoint pairs are disjoint.

Hence there are at least five covered points. Equality would give a proper five-edge-colouring of K_6, necessarily a one-factorization. After relabelling, three factors are

$$01|23|45, \qquad 02|14|35, \qquad 03|15|24.$$

Send ℓ to infinity and normalize the first two directions to horizontal and vertical. Put $p_0 = (0,0)$, $p_1 = (x,0)$ and $p_2 = (0,y)$, where $xy \neq 0$. Writing $p_3 = (a,y)$, the first two factors force $p_4 = (x,b)$ and $p_5 = (a,b)$. Parallelism of 03 with 15 and with 24 gives two determinant equations whose difference is $2xy = 0$, impossible for odd q. Thus a disjoint line has at least six covered points, and in all cases at most $(q+1) - 6 = q - 5$ points of $\mathcal{U}(A)$ remain. □

Theorem 4.3 (Rigidity theorem)**.** *For a six-arc $A \subset \mathrm{PG}(2,11)$, the following are equivalent:*

(i) the projective deep-hole syndrome locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);

(ii) $\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$-points of a nonsingular conic;

(iii) $|\mathcal{U}(A)| \leq 15$ (in fact, $|\mathcal{U}(A)| = 12$);

(iv) A is $\mathrm{PGL}(3,11)$-equivalent to the Clebsch hexagon;

(v) the stabilizer of A in $\mathrm{PGL}(3,11)$ contains A_5.

Proof. Suppose first that $\mathcal{U}(A)$ lies on a conic Q. If Q is nonsingular then $|\mathcal{U}(A)| \leq q+1 = 12$. If Q is degenerate, its rational points lie on one or two rational lines when it splits; in the nonsplit rank-two case its only rational point is the singular point. Thus Lemma 4.2 again gives $|\mathcal{U}(A)| \leq 12$. On the other hand, the chord-defect identity of Lemma 6.2 says

$$|\mathcal{U}(A)| = 22 - c(A),$$

where $c(A)$ is the number of nonvertex triple-edge concurrences. In Dye's terminology these are precisely the Brianchon points of the hexagon A. Over a field of characteristic different from two, Dye proves that a hexagon has at most ten Brianchon points and that the equality cases form a single $\mathrm{PGL}_3(K)$-orbit [14, §2.2–2.3, Theorem 1, pp. 275–276]. Hence $c(A) \leq 10$, so $|\mathcal{U}(A)| \geq 12$; equality holds throughout, $c(A) = 10$, and A is a Clebsch hexagon. This proves (i)⇒(iv). Proposition 3.2 and projective equivariance give (iv)⇒(ii)⇒(i). Dye's stabilizer theorem gives (iv)⇒(v), and Storme and Van Maldeghem's finite-plane classification gives the converse.

It remains only to connect the numerical condition (iii). Normalize a projective frame in every six-arc and sweep the two remaining vertices. This gives 1548 representatives meeting every projective class. Exact secant enumeration yields

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\}.$$

The corresponding representative multiplicities are

$$(6, 30, 150, 300, 630, 360, 72).$$

The six representatives attaining 12 form the single Clebsch orbit ($6 = 360/60$); every other class has at least 16 extension points. Thus (iii)⇔(iv), completing the equivalence. The exhaustive calculation is therefore used for the size-gap clause, not for the conic- containment implication. □

The standalone artifact `check_rigidity_degenerate_conic.py` implements the exhaustive computation just described, reproduces the displayed histogram, and returns exactly the six normalized Clebsch-class representatives whose extension loci lie on conics; every containing conic is nonsingular.

The same census supports a strengthening in which a cubic, rather than a conic, is allowed.

Proposition 4.4 (Low-degree algebraic rigidity)**.** *For a six-arc $A \subset \mathrm{PG}(2, 11)$, the uncovered locus $\mathcal{U}(A)$ is contained in the zero set of a nonzero homogeneous form of degree at most three if and only if A is projectively equivalent to the Clebsch hexagon. In the Clebsch case the least such degree is two: the quadratic vanishing space is spanned by the defining nonsingular conic equation Q, and the cubic vanishing space is*

$$Q \cdot \mathbb{F}_{11}[X, Y, Z]_1.$$

Computer-assisted proof. For each of the fifteen projective classes in the six-arc census, evaluate the ten cubic monomials on $\mathcal{U}(A)$. The resulting evaluation map has rank ten for every non-Clebsch class and rank seven for the Clebsch class. Thus no nonzero cubic vanishes on a non-Clebsch uncovered locus. A containing form of smaller positive degree could be multiplied by a nonzero monomial to produce such a cubic, so smaller positive degrees are excluded as well; no nonzero constant vanishes on the nonempty locus. For the Clebsch class, the quadratic kernel is one-dimensional and generated by Q; multiplication by the three-dimensional space of linear forms gives the full three-dimensional cubic kernel. These ranks and kernel identities are recomputed exactly by `check_low_degree_loci.py`. □

Corollary 4.5 (Monomial characterization)**.** *Up to monomial equivalence, C is the unique linear $[6, 3, 4]_{11}$ code of covering radius three whose projective deep-hole syndrome locus is contained in a plane curve of degree at most three. In particular, A_5 is recovered from this coding condition, not assumed as a hypothesis.*

Proof. The columns of a parity-check matrix of such a code form a six-arc A, and covering radius three identifies its projective deep-hole syndrome locus with $\mathcal{U}(A)$. Proposition 4.4 makes A projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. The converse is immediate, and the stabilizer statement is clause (v) of the theorem. □

Remark 4.6 (Sharpness of the degree threshold). Degree three is best possible. Apart from the Clebsch conic, the exhaustive search through degree five finds precisely two projective classes whose uncovered locus is the full set of $\mathbb{F}_{11}$-rational zeros of a form: one 18-point locus is a smooth absolutely irreducible quartic of genus three, and one 19-point locus is an absolutely irreducible quintic with one ordinary nonsplit node, whose normalization has genus five. The artifact also finds sixteen exact sextics in the first nonzero kernel of one 22-point class, including a smooth absolutely irreducible witness. We make no degree-six classification: the larger degree-six kernels of the other classes were not exhausted.

Remark 4.7 (Which implications are new). The equivalence (iv)⇔(v) is not ours and is not new: that the Clebsch hexagon is the unique $\mathrm{PGL}(3, 11)$-orbit of six-arcs whose stabilizer contains A_5 is Storme and Van Maldeghem [25, Prop. 12], quoted in Section 2 above, and the classification of six-arcs that underlies it is Sadeh's. The content of the theorem is the implication (i)$\Rightarrow$(iv), and with it the equivalence of the projective syndrome conditions (i)–(iii) with the classical characterizations (iv)–(v). It is that bridge — from a covering-radius property of the code to a projective classification — that is the claim, and the corollary above is its consequence.

The rigidity theorem has three quantitative refinements: a global gap in the extension-count census, a projectively invariant distance from the nearest conic, and a local gap under single-vertex replacement. The first is a statement of the standard extremal shape — there is no object strictly between the extremal value and the next, and the objects attaining the extremal value are classified — familiar from the minimum-side gap results for arcs and blocking sets, such as Blokhuis–Bruen's [12] theorem on the sizes realized just above the minimum. Here the invariant is the projective deep-hole-direction count $|\mathcal{U}(A)|$.

For the second, let $\mathfrak{C}$ be the set of nonsingular conics in $\mathrm{PG}(2,11)$ and define the *nearest-conic discrepancy*

$$\delta(A) = \min_{\mathcal{Q}\in\mathfrak{C}} |\mathcal{U}(A) \triangle \mathcal{Q}(\mathbb{F}_{11})|.$$

This is a PGL-invariant function, not a metric on arcs: equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A)$, while PGL permutes $\mathfrak{C}$.

For the local statement, let $\mathcal{G}_1$ be the graph whose vertices are embedded six-arcs in $\mathrm{PG}(2,11)$, with two vertices adjacent when their point sets have symmetric difference two. Fix the displayed Clebsch arc A and its standard conic $\mathcal{C}$, and put

$$d_{\mathcal{C}}(B) = |\mathcal{U}(B) \triangle \mathcal{C}|$$

for every embedded six-arc B; clause (c) restricts this discrepancy to the neighbors of A.

Theorem 4.8 (Gap theorem)**.** *The Clebsch configuration is rigid rather than merely stable.*

(a) (Census gap.) *There is no six-arc $A \subset \mathrm{PG}(2,11)$ with $12 < |\mathcal{U}(A)| < 16$. Furthermore, every six-arc attaining $|\mathcal{U}(A)| = 12$ is $\mathrm{PGL}(3,11)$-equivalent to the Clebsch hexagon, and every six-arc with $|\mathcal{U}(A)| \geq 16$ has $\mathcal{U}(A)$ contained in no conic.*

(b) (Global nearest-conic gap.) *On the fifteen $\mathrm{PGL}(3,11)$-classes of six-arcs, the exact class-counted histogram of δ is*

$$\{0{:}1,\ 12{:}2,\ 13{:}1,\ 14{:}3,\ 15{:}1,\ 16{:}4,\ 17{:}2,\ 18{:}1\}.$$

Its unique zero is the Clebsch class. Thus every non-Clebsch class has $\delta(A) \geq 12$, and the bound is sharp.

(c) (Perturbation gap.) *Each deleted vertex of A has exactly 42 legal replacements, giving 252 distinct neighbors B. Their exact fixed-conic discrepancy histogram is*

$$\{18{:}30,\ 19{:}60,\ 20{:}90,\ 22{:}42,\ 24{:}30\}.$$

The corresponding histogram of $|\mathcal{U}(B) \cap \mathcal{C}|$ is $\{4{:}60, 6{:}132, 7{:}60\}$, so at most seven conic directions remain; no neighbor has $\mathcal{U}(B)$ contained in any conic, degenerate or otherwise. Under $\mathrm{Stab}(A) \cong A_5$, the neighbors split into eight orbits whose pairs (orbit size, fixed-conic discrepancy) are

$$(30,18), (60,19), (30,20), (30,20), (30,20), (12,22), (30,22), (30,24).$$

Thus every vertex adjacent to A in $\mathcal{G}_1$ has $d_{\mathcal{C}}(B) \geq 18$.

Remark 4.9 (Global versus fixed-conic discrepancy). The closing statement is local to the fixed pair $(A, \mathcal{C})$ and is not a global nearest-arc theorem. The conic-preserving projectivity $g : (X, Y, Z) \mapsto (4X, 2Y, Z)$ sends A to a distinct embedded six-arc, while equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A) = \mathcal{C}$. Hence $d_{\mathcal{C}}(gA) = 0$, although $gA \neq A$. This is exactly why the global statement minimizes over conics and passes to projective classes: gA remains in the Clebsch class and has $\delta(gA) = 0$. The lower bound eighteen is specifically for the neighbors of the fixed embedded A in $\mathcal{G}_1$; the global non-Clebsch lower bound is the separate value twelve.

Remark 4.10 (Two spectra, one word apart). Clause (a) is a gap in the spectrum of the *projective deep-hole- direction count* $|\mathcal{U}(A)|$ at fixed arc size $k = 6$. It is not a statement about the much-studied spectrum of the *sizes* k for which a complete k-arc exists, which at $q = 11$ was determined by Faina, Marcugini, Milani and Pambianco [13]: there the invariant is k itself and the relevant constants are $t(2, 11) = 7$ and $m''(2, 11) = 10$. The two meet only at the boundary $\mathcal{U}(A) = \emptyset$, where an arc is complete; that $t(2, 11) = 7$ exceeds six is exactly why no six-arc of $\mathrm{PG}(2, 11)$ is complete, and hence why the histogram above has no zero bin.

Computer-assisted proof. The first clause restates the exhaustive classification of Theorem 4.3. For the second, `check_global_conic_gap.py` normalizes all 360 ordered four-frames of each of the 1548 representatives to obtain exactly fifteen projective-class keys. Independently it enumerates all 177156 projective ternary quadrics, retains exactly 160930 nonsingular conics, and computes the minimum symmetric difference for each class. It asserts the displayed class histogram, a direct-XOR/intersection-formula cross-check, and the unique zero class.

For the third clause, the local checker enumerates the 133 projective points, deletes each of the six vertices in turn, and tests every replacement outside A for the six-arc condition. It asserts the six per-vertex counts, distinctness of the 252 resulting arcs, both displayed histograms, and conic noncontainment by exact quadratic-monomial rank. It then uses the sixty exact projectivities in the stabilizer to certify the eight closed orbits and recover both histograms from their rows. It also verifies the projectivity in the preceding remark and directly recomputes $\mathcal{U}(gA) = \mathcal{C}$. Its source is `check_perturbation_gap.py`. □

All three clauses are exhaustive finite checks over $\mathrm{PG}(2, 11)$; the three standalone paper-package checkers expose their exact assertions and exit nonzero on any mismatch.

Remark 4.11 (Why these facts are not already recorded). The locus $\mathcal{U}(A)$ is not a new object: by Segre's theory of the tangent envelope of an arc, a point extends A exactly when its pencil is a component of that envelope, so $\mathcal{U}(A)$ is the classical extension locus — see Ball and Lavrauw [8, Thms. 39–41] for the statement in both parities. Two separate reasons account for the absence of the facts above from that literature, and neither is that the objects were unknown. First, the arc-classification line has long had the census we use — the six-arc classes of $\mathrm{PG}(2, 11)$ are Sadeh's — but no motive to compute $\mathcal{U}(A)$ on it: its purpose there is the construction of cubic surfaces, which consumes only the condition that A itself lie on no conic. Second, the arc-extension line has exactly this motive, but its theorems are large-k theorems whose hypotheses exclude our case: over $\mathbb{F}_q$ with q odd the envelope results apply from $|A| \geq 2q/3 + 2$, which is ten points at $q = 11$, and Segre's extension bound needs $k > q - \sqrt{q}/4 + 25/16 \approx 11.73$, against our $k = 6$. At $k = 6$ the envelope has class $14 > 12$, so membership in it constrains nothing. The interest of the statements above therefore rests not on their inaccessibility but on the coding reading of $\mathcal{U}(A)$ and on the rigidity theorem that the gap protects.

5 The chirality bipartition

The six columns of A give $\binom{6}{3} = 20$ three-element subsets. The stabilizer A_5 has two orbits of size ten on them, interchanged by complementation: for every S, the orbit of S never contains S^{c}. The triples also form ten complementary pairs $\{S, S^{\mathsf{c}}\}$. Choose the member in either one of the two A_5 orbits consistently from every pair and join two pairs when their chosen representatives share exactly two columns. The resulting graph is the Petersen graph; choosing the other orbit gives the same adjacency because simultaneous complementation preserves intersection sizes.

There is a direct geometric labelling of these ten vertices.

Proposition 5.1 (Brianchon–support dictionary)**.** *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [4, 14], regarded as the five perfect matchings in their synthematic total, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T, T')$ be the perfect matching joining opposite vertices of this cycle, and let $\beta(T, T') = \{S, S^{\mathsf{c}}\}$ be its unordered bipartition.*

The three chords indexed by $M(T, T')$ concur at a point $b(T, T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \qquad \{T, T'\} \longmapsto b(T, T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S, S^{\mathsf{c}}\} : |S| = 3\}, \qquad \{T, T'\} \longmapsto \beta(T, T'),$$

are A_5-equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of the synthematic total. Hence $\{T, T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T, T')$, so the ten triangle pairs give exactly the ten perfect matchings outside the total. Edge's Clebsch construction says that the three chords of each such matching concur, at the ten distinct Brianchon points [4, §§19–20,30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above. All concurrences, the ten multiplicity-three and fifteen multiplicity-one intersections of disjoint chords, and the equivariance and support-map compatibility follow from these two equivariant bijections. □

As an independent verification, `check_chirality.py` checks all chord intersections, the exact $3^{10}1^{15}$ multiplicity ledger, both equivariance statements, and compatibility with the support map.

Corollary 5.2 (The decoder reconstructs the Brianchon configuration)**.** *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five-matching synthematic total.*

Proof. The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 5.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings complementary to the total. □

Edge's two systems of eleven Clebsch hexagons over a fixed conic, exchanged by the operations outside $\Omega^+(3, 11)$ [4, §§29,32], are a classical antecedent of the same two-class motif. They are classes of hexagons, however, not the support-orbit bipartition proved below. Blokhuis–Seress–Wilbrink report another, distinct Petersen appearance at $q = 31$: a sixteen-point complete exterior set split into a six-arc and a ten-set; every 2-secant of the six-arc is also a 2-secant of the ten-set, and the resulting fifteen pairs of ten-set points are the Petersen edges [2, p. 146]. That configuration is neither the present $q = 11$ support graph nor a chirality assertion.

Proposition 5.3 (Invariant support chirality)**.** *The twenty three-element coordinate supports split into two complementary A_5-orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s, each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the 2400 minimum-weight leaders split as $1200 + 1200$.*

Every monomial automorphism of C preserves this unordered bipartition. The other coset in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. Thus the intrinsic structure is a two-class decomposition, or an unbased $\mathbb{Z}/2$-torsor, rather than a canonically oriented $\mathbb{Z}/2$-valued label. By Proposition 5.1, the ten complementary support pairs are canonically both the two-subsets of the five self-polar triangles and the ten Brianchon points.

Proof. The two size-ten support orbits and the normalizer action follow by applying the explicit exotic degree-six A_5 to the $\binom{6}{3} = 20$ subsets. Of the six synthematic totals on the six coordinates, exactly one is stabilized by this A_5 (indeed by its full S_5 normalizer); it is the total supplied by the five triangles. Any two of its perfect matchings are edge-disjoint, so their union is a connected 2-regular bipartite graph on six vertices, hence a six-cycle with a unique unordered $3+3$ bipartition. The ten pairs of matchings give all ten complementary support pairs. This construction is equivariant under the full normalizer, and direct comparison shows that disjoint pairs of triangles give exactly the intersection-two adjacency defined above. Fix a maximum-distance syndrome s and a three-support S. The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s. None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A, contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S. Finally, every monomial automorphism induces a support permutation in A_5, by the exact sequence in Section 3.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits, but the pure-permutation audit there and the monomial support-image audit show that it does not preserve C, even after coordinate rescaling. □

The complete finite ledger in the preceding proof is independently checked by `check_chirality.py`, including the support orbits, Petersen adjacency, all perfect matchings and synthematic totals, the triangle-pair/support/Brianchon dictionary, both equivariance statements, the full normalizer, all $120 \cdot 20$ coefficient-bearing leaders, and all $600 \cdot 2400$ monomial-action cases.

The bipartition is therefore intrinsic to the code's covering-radius data, but its two sides have no preferred signs without an additional orientation.

There is a finer equivariance distinction. The stabilizer in $\mathrm{MAut}(C)$ of one *affine* deep-hole syndrome has order five and four orbits of size five on its twenty leaders. Transporting any one orbit gives a monomial-equivariant set-valued decoder of constant list size five on the 120 deep-hole syndromes, and every nonempty equivariant decoder on this syndrome orbit has list size at least five. Thus chirality is not the minimum or coarsest arbitrary equivariant selection: its two size-ten lists are the two proper nonempty selections determined solely by the support, each a union of two of the size-five orbits. These affine stabilizer orbits, their transport, and the resulting 288000 equivariance cases are checked by `check_decoding.py`; no orientation of the two chirality classes is chosen.

6 Why $q = 11$

The conic-filling syndrome phenomenon of Sections 3–4 is specific to $q = 11$, even without a symmetry hypothesis on the arc. A universal chord-multiplicity identity makes this a four-field question, and the only nontrivial residue is a classical exact-distance problem in the Sylvester graph.

Lemma 6.1 (Secant-covering bound)**.** *Let A be a six-arc of $\mathrm{PG}(2, q)$ disjoint from a nonsingular conic $\mathcal{C}$, and suppose A is complete outside $\mathcal{C}$: every point of $\mathrm{PG}(2, q)$ lying off A and off $\mathcal{C}$ lies on some secant of A. Then*

$$15(q - 1) \;\geq\; q^2 - 6,$$

equivalently $q^2 - 15q + 9 \leq 0$, so $q \leq 14$.

Proof. Since A has six points and $\mathcal{C}$ has $q + 1$ points, and A is disjoint from $\mathcal{C}$,

$$|\mathrm{PG}(2, q) \setminus (A \cup \mathcal{C})| = (q^2 + q + 1) - 6 - (q + 1) = q^2 - 6.$$

Each of the $\binom{6}{2} = 15$ secants of A is a line of $\mathrm{PG}(2, q)$, hence has $q+1$ points, exactly two of which lie in A; so each secant covers at most $q - 1$ points of $\mathrm{PG}(2, q) \setminus A$, and in particular at most $q - 1$ points of $\mathrm{PG}(2, q) \setminus (A \cup \mathcal{C})$. Completeness outside $\mathcal{C}$ says the fifteen secants together cover all $q^2 - 6$ points of this set, so

$$q^2 - 6 \;\leq\; 15(q - 1).$$

(No accounting for overlap between secants is needed here: the bound only uses that fifteen sets of size at most $q - 1$ cover a set of size $q^2 - 6$, which requires the sum of their sizes to be at least $q^2 - 6$ regardless of how much they overlap; overlap can only make the true covered count smaller than the sum, never larger, so the inequality direction needed for an upper bound on q is exactly the one overlap cannot violate.) Rearranging, $q^2 - 15q + 9 \leq 0$, whose real roots are $\frac{15 \pm \sqrt{189}}{2} \approx 0.62, 14.38$; since q is a positive integer, $q \leq 14$. □

Lemma 6.2 (Chord-defect identity)**.** *Let A be a six-arc in $\mathrm{PG}(2, q)$, and let $c(A)$ be the number of points off A incident with three of its fifteen chords. Then*

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \qquad 0 \leq c(A) \leq 15.$$

Consequently, if $|\mathcal{U}(A)| = q + 1$, then

$$c(A) = (q - 6)(q - 9).$$

Proof. At a point off A, concurrent chords have pairwise disjoint endpoint pairs, so at most three chords occur. Let n_i count the off-arc points on exactly i chords. Counting chord–point incidences and pairs of chords with disjoint endpoints gives

$$n_1 + 2n_2 + 3n_3 = 15(q - 1), \qquad n_2 + 3n_3 = \binom{15}{2} - 6\binom{5}{2} = 45.$$

Writing $c(A) = n_3$, the number of covered points off A is $n_1 + n_2 + n_3 = 15q - 60 + c(A)$. Subtracting this from the q^2+q-5 points off A proves the identity. Each triple concurrence uses one of the fifteen perfect matchings of the six endpoints, and a fixed matching can be concurrent at only one point; hence $c(A) \leq 15$. The last display follows on setting $|\mathcal{U}(A)| = q + 1$. $\square$

Theorem 6.3 (Uniqueness of $q = 11$)**.** *Let q be a prime power. If a six-arc $A \subset \mathrm{PG}(2, q)$ satisfies $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{C}$, then $q = 11$. Moreover, at $q = 11$ every such A is projectively equivalent to the Clebsch hexagon.*

Proof. Lemma 6.2 gives $c(A) = (q - 6)(q - 9)$ with $0 \leq c(A) \leq 15$. The prime-power condition therefore leaves only $q \in \{4, 5, 9, 11\}$: the values at 7 and 8 are negative, while every integer $q \geq 12$ gives $c(A) \geq 18$.

At $q = 4$, every six-arc is a hyperoval. Each point off it lies on three secants (count the six hyperoval points on the five lines through the point), so $\mathcal{U}(A)$ is empty. At $q = 5$, every six-arc is an oval and hence a nonsingular conic; the standard internal/external point counts show that every off-conic point lies on a secant, so again $\mathcal{U}(A)$ is empty.

It remains to exclude $q = 9$. Relative to the hypothetical conic $\mathcal{C}$, every chord of A is passant. A point off $\mathcal{C}$ lies on five passants if it is internal and four if it is external; hence the five chords through each vertex force all six vertices of A to be internal. Let H be the graph on the 36 internal points, with $P \sim Q$ when $Q \in P^\perp$ under the conic polarity. Standard conic counts give the intersection array

$$\{5, 4, 2; 1, 1, 4\},$$

so H is the Sylvester graph [10, §13.1.2]; see also the combinatorial uniqueness proof in [11, Thm. 6]. For distinct internal points, the unique possible common neighbor is $P^\perp \cap Q^\perp = (PQ)^\perp$; consequently PQ is passant exactly when P, Q are at distance two in H. Thus A would be a six-clique in the exact-distance-two graph of H. But [9, Table 1] gives $\mathrm{eq}_2(H) = 5$, a contradiction.

Therefore $q = 11$. The final assertion follows from Theorem 4.3, which puts every matching $q = 11$ representative in the single Clebsch orbit. $\square$

The smaller from-scratch certificate `../../notes/2026-07-15-c181-sylvester-q9.py` independently constructs $\mathbb{F}_9$, $\mathrm{PG}(2, 9)$, the polarity graph and its full intersection array, verifies passant join if and only if distance two for all internal-point pairs, and computes the exact clique number 5. The all-field checker `check_small_q_uniqueness.py` supplies the separate census below.

For independent verification, every six-arc contains an ordered projective four-frame, which may be normalized to $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$. Enumerating the remaining unordered pair with exact field arithmetic gives the following census; exponents in the third column record multiplicities.

q	representatives	histogram of $\lvert\mathcal{U}(A)\rvert$	conic matches
2	0	—	0
3	0	—	0
4	1	0^1	0
5	3	0^3	0
7	70	$0^{40}, 2^{30}$	0
8	195	$0^{45}, 4^{150}$	0
9	441	$0^6, 4^{15}, 6^{120}, 7^{120}, 8^{180}$	0
11	1548	$12^6, 16^{30}, 18^{150}, 19^{300}, 20^{630}, 21^{360}, 22^{72}$	6
13	4015	$36^{85}, 38^{210}, 39^{480}, 40^{1080}, 41^{1800}, 42^{360}$	0

The enumeration uses exact field arithmetic, including explicit asserted tables for $\mathbb{F}_4, \mathbb{F}_8$, and $\mathbb{F}_9$; its conic test is also exercised on nonsingular and singular forms in characteristic two. It is reproduced by `check_small_q_uniqueness.py` and agrees with the conceptual proof above.

Proposition 6.4 (The Clebsch-family uncovered formula). *Let H be a Clebsch hexagon over $\mathbb{F}_q$. Then*

$$|\mathcal{U}(H)| = q^2 - 14q + 45.$$

If $q \equiv 3 \pmod 4$ and $\mathcal{C}$ is Dye's associated conic, then $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(H)$ and

$$|\mathcal{U}(H) \setminus \mathcal{C}(\mathbb{F}_q)| = (q-4)(q-11).$$

In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if $q = 11$.

Proof. Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [14, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by $c(H)$. Thus $c(H) = 10$, and Lemma 6.2 gives

$$|\mathcal{U}(H)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of H are non-secants of the associated conic when $q \equiv 3 \pmod 4$[14, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of $\mathcal{C}$ is uncovered. Subtracting $|\mathcal{C}(\mathbb{F}_q)| = q + 1$ from the first formula gives

$$q^2 - 15q + 44 = (q-4)(q-11).$$

The factor vanishes only at $q = 4$ and $q = 11$, and the stated congruence leaves only $q = 11$. Conversely, Clebsch hexagons exist over $\mathbb{F}_{11}$ by Dye's Theorem 1, and the inclusion above is then an equality because both sets have twelve points. □

The theorem is an unconditional field-size classification with a conceptual small-field proof. Within the icosahedral construction, three further features meet at 11: $p + 1 = 12$, so the twelve vertices of the icosahedron exhaust the projective line; p is prime to 60, so the icosahedral group survives reduction faithfully; and the fifteen secants of a six-arc have enough capacity to cover every point off the conic. The failure of the same symmetric recipe at the next admissible prime is nevertheless worth recording concretely. At $q = 19$ ($19 \equiv -1 \pmod{10}$, so the rationality filter passes and only Lemma 6.1 excludes it) the same recipe — the poles of the six five-fold axes of an icosahedral $A_5 < \mathrm{PSL}_2(19)$ — does still produce a genuine six-arc disjoint from the conic, so the object exists; what fails is the covering. Proposition 6.4 now gives directly

$$|\mathcal{U}(A)| = 140 = 20 + 120$$

against a conic of only $q+1 = 20$ points. The checker `check_q19_nonexample.py` independently reconstructs the arc and reproduces this value. The arithmetic differs from $q = 11$ in a way worth noting: there $5 \mid q-1$, so an order-five element splits and fixes two rational points of $\mathcal{C}$, whereas at $q = 19$ we have $5 \mid q+1$, so the order-five elements lie in a non-split torus and fix no rational conic point at all — their fixed pairs are conjugate over $\mathbb{F}_{361}$. The chord through such a pair is Galois-stable and hence still a rational line, so its pole is still a rational point and the arc is still defined; we verify this in the script rather than assume it.

The shape of the failure is instructive. The twenty conic directions remain in the projective deep-hole syndrome locus — $\mathcal{C} \subset \mathcal{U}(A)$ still — by the same classical edge criterion [14, p. 281], but they no longer exhaust $\mathcal{U}(A)$: $120 = (19-4)(19-11)$ further points of $\mathrm{PG}(2,19)$ escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic (the quadratic-monomial evaluation matrix of $\mathcal{U}(A)$ has full rank six). The earlier capacity bound from Lemma 6.1 predicts only that at least 85 off-conic points escape; the Clebsch-family formula gives the exact excess 120. At $q = 11$ the same capacity, 150, comfortably exceeds the 115 points needing cover, and the excess is spent on overlap. The Clebsch objects therefore persist, and the associated conic remains inside the projective syndrome locus when $q \equiv 3 \pmod 4$, but the exact filling is isolated at $q = 11$ by the factorization above.

6.1 The small-arc boundary

The chord moments also put the six-point theorem into a short classification for all smaller neighboring arc sizes.

Theorem 6.5 (Conic-filling loci for $4 \le k \le 7$)**.** *Let A be a k-arc in $\mathrm{PG}(2,q)$, where q is a prime power and $4 \le k \le 7$. If $\mathcal{U}(A)$ is the full set of $\mathbb{F}_q$-rational points of a nonsingular conic, then*

$$(k,q) = (4,5) \qquad \text{or} \qquad (k,q) = (6,11).$$

In the first case A is a projective four-frame; in the second it is a Clebsch hexagon. Conversely, both cases occur.

Proof with exhaustive finite verification. Let n_i count off-arc points incident with exactly i chords. For $k \le 7$, at most three pairwise disjoint chords concur. Counting chord–point incidences and pairs of disjoint chords gives the two universal moments

$$\sum_i i n_i = \binom{k}{2}(q-1), \qquad \sum_i \binom{i}{2} n_i = 3\binom{k}{4}.$$

For $k = 4$, the second moment forces $n_2 = 3$, and hence $|\mathcal{U}(A)| = (q-2)(q-3)$. Equating this with $q+1$ gives $q = 5$. Every four-arc is a projective frame, and for the standard frame a direct substitution gives

$$\mathcal{U}(A) = Z(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

a nonsingular six-point conic. For $k = 5$, the same moments force $n_2 = 15$ and $|\mathcal{U}(A)| = q^2 - 9q + 21$; equality with $q + 1$ would require $q^2 - 10q + 20 = 0$, which has no integral root. The case $k = 6$ is Theorem 6.3 together with Theorem 4.3.

For $k = 7$, the moments give

$$|\mathcal{U}(A)| = q^2 - 20q + 120 - n_3.$$

If this equals $q + 1$, then

$$n_3 = q^2 - 21q + 119, \quad n_2 = 105 - 3n_3, \quad n_1 = 3(q^2 - 14q + 42).$$

The standard arc-size bound gives $q \ge 7$. The condition $n_2 \ge 0$ gives $q \le 15$, so the prime-power candidates are $q \in \{7, 8, 9, 11, 13\}$; then $n_1 \ge 0$ eliminates $7, 8, 9$. Thus only $q = 11$ and $q = 13$ remain, with forced spectra $(n_1, n_2, n_3) = (27, 78, 9)$ and $(87, 60, 15)$ respectively. An exhaustive four-frame-normalized enumeration finds no conic-filling seven-arc in either plane. Every seven-arc is reached because it contains a four-frame; equivalently, the checker extends every normalized six-arc by every point of its uncovered locus and verifies that each resulting seven-arc occurs exactly three times, according to which non-frame point is deleted. For all candidates with $|\mathcal{U}(A)| = q+1$, the quadratic evaluation matrix has full rank six. The complete finite leaves, including the $q = 5$ equality, are replayed by `check_small_k_conic_filling.py`; the universal moment reductions are kernel-checked in `SmallKChordMoments.lean` and `SmallKGeometricBridge.lean` under `lean/RelativeConicArcs/`. □

The $q = 5$ four-frame equality is elementary and is stated here without an independent priority claim. Dye's adjacent $q = 5$ degeneration concerns the six points on a conic and their ten internal triple-chord points, not this four-frame uncovered-locus statement [14, §1.4]. The role of the theorem here is to give the unified range $4 \leq k \leq 7$; its new certified component is the exclusion of both surviving seven-arc fields. For $k \geq 8$ the second moment leaves additional concurrency parameters, so the analogous classification remains open.

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